Mercy House

Can a detective establish the truth when loyalty and plausible evidence point in the wrong direction?

Task and world

A fictional mountain town, 1900. Sixteen profiles, fourteen places. The detective receives its remembered life and the local crime scene, then chooses where to go, whom to question and what to inspect. A recorded LLM incident tape is executed at seeded reset; the injury and planted threat create causal traces. Historical background is authored. Post-reset characters use isolated private memories and live model calls.

Task / level	Objective
last_witness	Fatal assault / introductory demo
borrowed_signature	Document fabrication / intermediate
brothers_letter	Concealment and recovery / advanced
garden_burials	Recovered remains and relocation / advanced
mercy_inquest	Seven material acts / advanced

Action and observation

Observe, identify people here, travel, search or delegate, secure material, ask with optional evidence, seek access, compare, excavate and report. Actions validate local presence, permissions, custody and types. Hearsay and partial perception retain their basis. A statement cannot create an object or permission. A real isolated shell can use the same public town tools.

Reward

```
R = clamp(C − 0.20 F − 0.10 U, 0, 1)
```

C is supported scoped target coverage; F and U are distinct false and unsupported findings. Full success requires C=1 and no F/U. A correct name alone gets zero. Duplicate findings cannot multiply credit. Programmatic proof requires acquired provenance; no LLM judge. Narrative is retained but unscored.

Verification, isolation and evidence

Defences

The protected world owns truth and reward. The shell has no repository, verifier, private logs, credential or orchestration token, and no external route. A trusted public gateway accepts only validated town actions. A separate model gateway holds keys and fixed endpoint configuration. All roles are unprivileged; shell filesystems, resources and subprocesses are bounded. Private planners receive deterministic question topics; public speakers see only released memories.

Run contract

Selected-task reference/no-op gates precede the detective. Same task/seed gives the same starting hash. MODEL/API_KEY/BASE_URL, TASK_ID, SEED, MODE and exact APPEND_FILE suffix are wired. Forty replies, ten executed calls per reply, six hundred seconds, tool deadlines and forced-end grading. Malformed responses consume a step and receive feedback. Public reply logs include raw input/output, actions, results, errors, hash, usage and reward; private audits remain protected.

Measured evidence

All five references score 1.0 and no-ops 0.0. 86 current town tests and 9 historical regression tests pass. The command isolation probe passes. A new no-cache builder produced all four images in 13.474 seconds locally.

Optional CPU RL: 20,000 episodes, 530.243 seconds, 460 coefficients, zero model calls. Held-out greedy 147/160 versus untrained 1/160. Training uses fixture NPCs and public candidate rules; LLM weights stay fixed. 2/5 completed live checkpoint exercises; small unpaired smoke, not whole-case success.

Known weaknesses

One canonical plot with five task scopes; small checkpoint families; authored proof vocabulary; live refusal and proposal distribution shift; no demonstrated complete-case LLM success. No direct score for internal emotions. Continuous offstage lives and a scored ghost mechanism are unfinished. Docker isolation assumes a trusted host/daemon. Organizer-machine timing and independent NPC deployment remain external checks. We make no universal security or winning claim.

Source requirements: supplied 2une.ai PS. Research: TextWorld (2018), Generative Agents (2023), Werewolf RL (ICML 2024), MACHIAVELLI (ICML 2023). Full primary links and competition-result caveats are in docs/RESEARCH_AND_WIN_PLAN.md.